

Bowen Song

Advanced Modeling / LLM / Ranking

Research engineer building ranking and reranking systems that survive strong baselines and real traffic.

Modern models need real evaluation, so I build ranking and reranking systems that survive strong baselines and traffic shifts.

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EXPERIENCE

PhD Research Intern, Core Ranking

2024

eBay | San Jose, CA

Ranking gains had to survive real traffic and strong baselines, so I built production-constrained model and evaluation workflows that made improvements worth trusting.

- Multi-objective ranking needed more selective capacity, so I designed Transformer MoE models that fused text and structured signals and created a scalable path to specialization.
- Traffic heterogeneity kept breaking naive expert routing, so I built distributed training and evaluation loops that preserved specialization gains under capacity limits and shifting demand.

Graduate Researcher, Personalized AI for Health

2020-Present

University of Southern California | Los Angeles, CA

Wearable sensing is noisy and deployment is constrained, so I built multimodal pipelines and personalized models that stay interpretable on device and useful across users.

- Wearable force estimation needed a smaller baseline, so I led SER and showed low-rank models could match or beat heavier LSTMs while staying interpretable and credible for edge deployment.
- Research pipelines were fragmented across sensing modalities, so I built end-to-end data, training, and evaluation workflows that made comparison under noise and subject shift reliable enough to guide model decisions.

Software Engineer, OpenShift

2019-2020

Red Hat (IBM) | Boston, MA

Production clusters need upgrades and debugging to be boring, so I shipped Kubernetes platform features and CI tooling that made operations safer and easier to trust.

- Ecosystem integration slowed teams adopting operators, so I led Helm and Operator Framework integration and widened the path for interoperability across tools.

PROJECTS

COT + RANKING

CoT-Rec

Chain-of-thought generation is noisy, so I turned reasoning traces into ranking candidates and improved final selection with reranking instead of trusting decode order.

SER PAPER IMPLEMENTATION

SER

Wearable GRF estimation needed a lighter baseline, so I implemented SER and showed interpretable low-rank modeling could stay accurate enough for deployment-minded biomechanics work.

SKILLS

Modeling: Transformers, ranking, mixture-of-experts, reranking, and evaluation design for selection-heavy models

ML Platform: PyTorch, TensorFlow, distributed training, experiment pipelines, and GPU clusters for production-minded ML; Krylov, Ray, Kubeflow, and evaluation loops for large-scale production ML iteration

PUBLICATIONS

Estimating Ground Reaction Forces From Inertial Sensors

IEEE TBME 2024 | B. Song, M. Paolieri, H. E. Stewart, L. Golubchik, J. L. McNitt-Gray, V. Misra, D. Shah

Sequence models are not always right, and this paper shows honest evaluation can favor a smaller architecture with clearer behavior.

Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements

IEEE CLOUD 2022 | Bowen Song, Marco Paolieri, Leana Golubchik

Optimization alone is not enough, and this paper connects performance analysis to incentives and the tradeoffs operators actually face.

EDUCATION

PhD in Computer Science

University of Southern California | Los Angeles, CA | 2020-2026 (expected)